Geometry

Section 1: Angle Relationships & Polygons

1 Two angles are complementary. If the measure of one angle is 14° more than three times the other, what is the measure of the smaller angle?

a. 19°

b. 24°

c. 38°

d. 71°

2 What is the sum of the interior angle measures of a convex hexagon?

a. 540°

b. 720°

c. 900°

d. 1080°

3 In $\triangle ABC$, the exterior angle at vertex C measures 125°. If one remote interior angle measures 55°, what is the measure of the other remote interior angle?

a. 55°

b. 65°

c. 70°

d. 75°

Section 2: Triangles, Similarity & Special Right Triangles

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4 In a right triangle, the legs have lengths 9 and 12. What is the altitude drawn to the hypotenuse?

a. 6.0

b. 7.2

c. 7.5

d. 8.0

5 In an isosceles right triangle (45°-45°-90°), the hypotenuse is $14\sqrt{2}$ cm. What is the perimeter of the triangle?

a. $28 + 14\sqrt{2}$ cm

b. $28\sqrt{2}$ cm

c. 42 cm

d. $14 + 14\sqrt{2}$ cm

6 Two similar triangles have perimeters of 24 cm and 36 cm. If the area of the smaller triangle is 32 cm^2, what is the area of the larger triangle?

a. 48 cm^2

b. 64 cm^2

c. 72 cm^2

d. 96 cm^2

Section 3: Circles, Arcs & Tangents

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7 A chord of length 16 cm is at a distance of 6 cm from the center of a circle. What is the radius of the circle?

a. 8 cm

b. 10 cm

c. 12 cm

d. 14 cm

8 What is the area of a circle whose circumference is 16π?

a. 16π

b. 32π

c. 64π

d. 128π

9 A central angle of 45° intercepts an arc in a circle with radius 8. What is the length of the intercepted arc?

a. π

b. 2π

c. 4π

d. 8π

Section 4: 3D Surface Area & Volume

10 What is the total volume of a right circular cylinder with radius 5 cm and height 12 cm?

a. 60π cm^3

b. 120π cm^3

c. 300π cm^3

d. 600π cm^3

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11 A rectangular water tank is 4 meters long, 3 meters wide, and 2 meters deep. If it is currently filled to 75% of its capacity, how many cubic meters of water are in the tank?

a. 12

b. 16

c. 18

d. 24

12 What is the surface area of a sphere with a radius of 6 cm?

a. 72π cm^2

b. 144π cm^2

c. 288π cm^2

d. 576π cm^2

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